BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

RESEARCH / PROTOTYPE SYSTEM DISCLAIMER: This system is a research/prototype fingerprint-based blood-group classification system. It is not clinically validated and must not be used as a substitute for standard serological blood-group testing or for medical/transfusion decisions. All predictions and advisory indicators are provided solely for evaluation purposes.

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000595	**Timestamp:**	2026-09-08 21:41:13
Input Source:	Mantra MFS100 Scanner | Standard Finger: LEFT THUMB	**Evaluation Status:**	**NORMAL**
Model Name:	Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%)	**Confidence:**	**95.52%**
Standard Acquisition Finger:	**LEFT THUMB** (Convention-based protocol — not automatically verified by scanner)		
Advisory Status:	**CONFIDENCE OK** (τ=70%)	**Advisory Reason:**	**NONE**

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



Predicted Blood Group:

O+

Confidence Score: **95.52%**

Advisory Threshold: **70%**

Advisory Screening: **CONFIDENCE OK**

Derived ABO Group: **O** (95.9% prob) | Rh Factor: **+**

RESEARCH ADVISORY: Advisory threshold passed (≥ 70.0%). Note: Confidence OK does not constitute medical verification. Standard serological testing required for medical/transfusion use.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
1.4%	1.5%	0.0%	1.1%	0.0%	0.0%	95.5%	0.4%